GOLDLITE SUPPLIES

KCBE GRADE 10 QUICK REVISION SERIES 2026

CORE MATHEMATICS

SERIES 1-10

SERIES 1

SECTION A (30 MARKS)

Answer ALL questions in this section.

1. The diagrams below represent common road signs:

(a)

(b)

State the number of lines of symmetry for the

Triangle: _________________ Octagon: _______________________

2. Given that A’ (3, -3) is the image of A (-1, -5) under a reflection. Find the equation of the mirror line in the form of ax+by+c= 0. (5 marks)

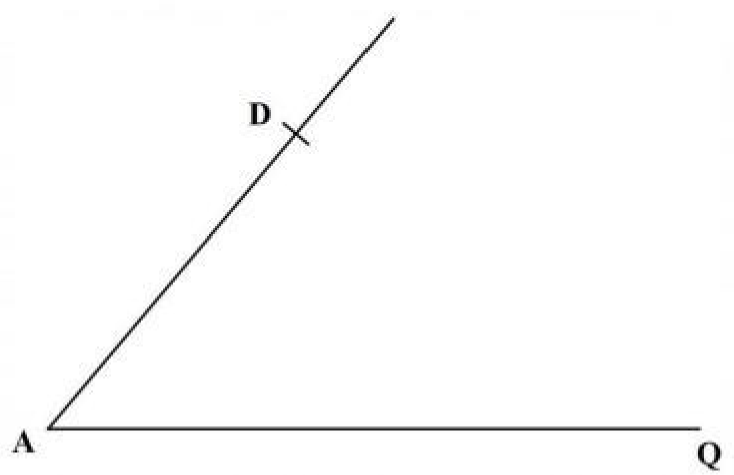
3. A straight line L_1 whose equation is $y = 2 - \frac{1}{3}x$ meets the y – axis at Q. Another straight line L_2 is perpendicular to L_1 at Q. Find the equation of L_2 in the form $y = mx + c$ where m and c are constants. (3 marks)

4. Solve for x in the equation $25^{x}=125^{2/3} \div 5^{-1}$(3 marks)

5. A carpenter had two big pieces of wood of equal length. The carpenter cut the first piece into smaller pieces of length 15 cm each without remainder. The carpenter cut the second piece into smaller pieces of length 24 cm each without remainder. Determine the minimum length of each of the big pieces of wood. (2 marks)

6. A cylindrical container of internal radius 10.5 cm has a hemispherical base. The container has water up to a height of 30.5 cm. Calculate the surface area of the container that is in contact with water. (Take $\pi = {}^{22}/_{7}$) (4 marks)

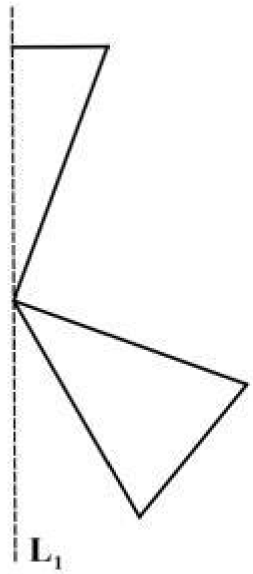
7. In the following figure, points A and D are vertices of a trapezium ABCD. Line DC is parallel to line AQ. Line DC = 4 cm. Point B lies on the line AQ such that angle DCB = 90°.



Using a rule and a pair of compasses only, complete the trapezium. Hence, measure the length of line AB. (4 marks)

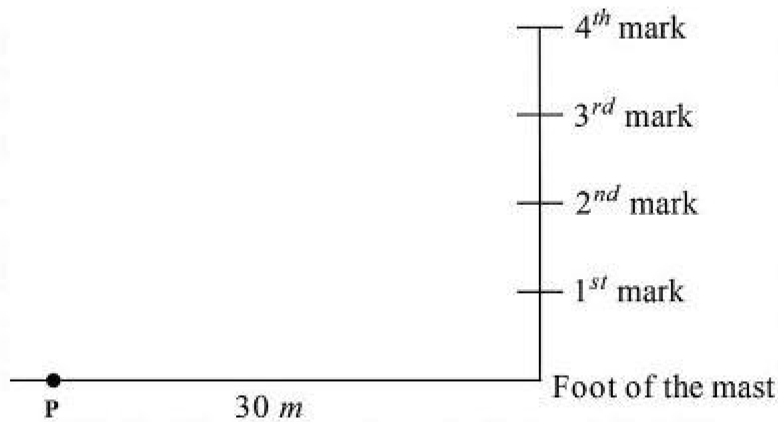
8. The length of a minor arc AB of a circle centre O is 10 cm. The arc AB subtends an angle of 1.25 radians at O. Calculate the area of the minor sector AOB. (3 marks)

9. The following figure represents part of a pattern. A line of symmetry, L_1, of the pattern is also shown.



Complete the pattern and hence state the order of rotational symmetry of the patterns. (3 marks)

10. The following figure (not drawn to scale) represents a communication mast. The mast has been divided into 4 equal parts. A point P is 30 m from the foot of the mast on the same ground level.



The angle of elevation of the 3rd mark from P is 50°. Calculate the height of the mast. (3 marks)

11. The following marks were scored by Grade 10 learners in a Mathematics quiz:

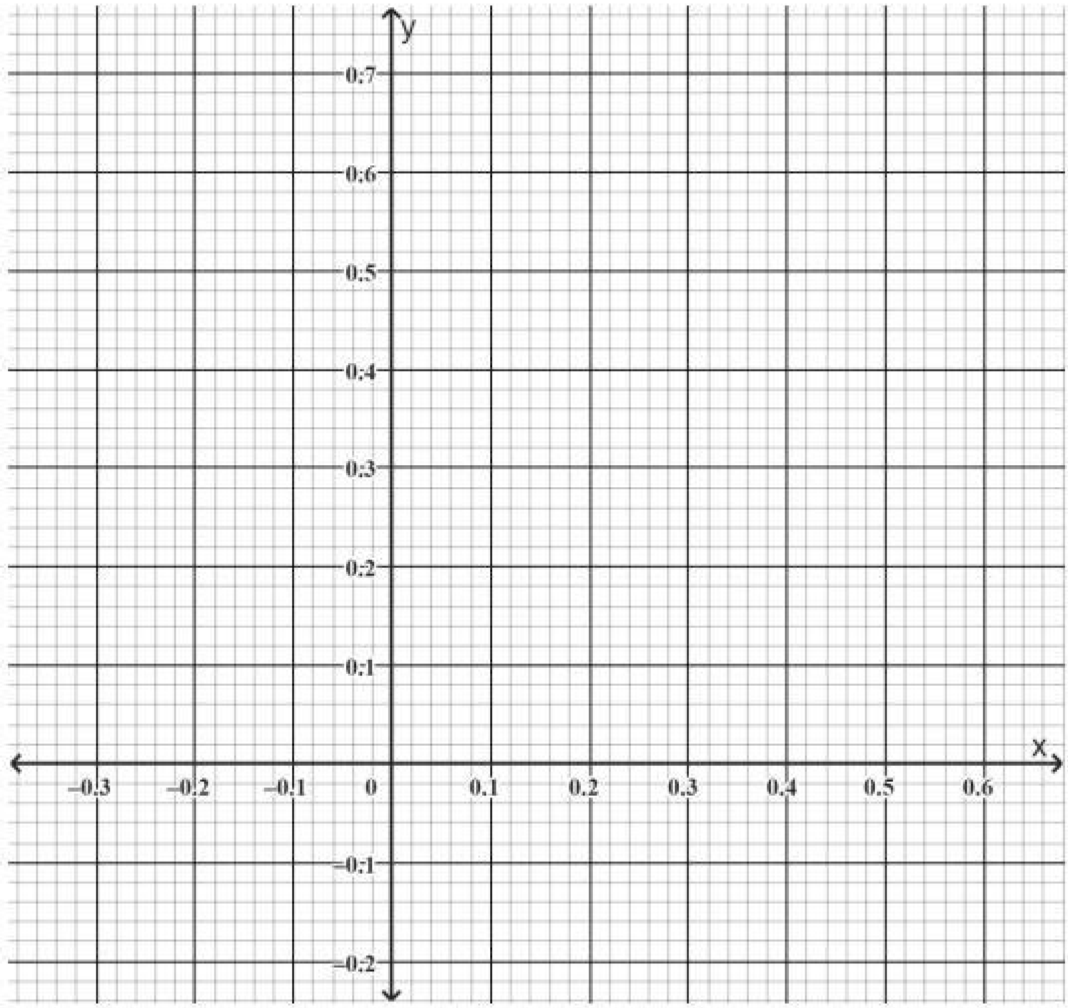
12, 15, 18, 12, 20, 15, 14, 12, 16

(a) Calculate the mean mark. (1 mark)

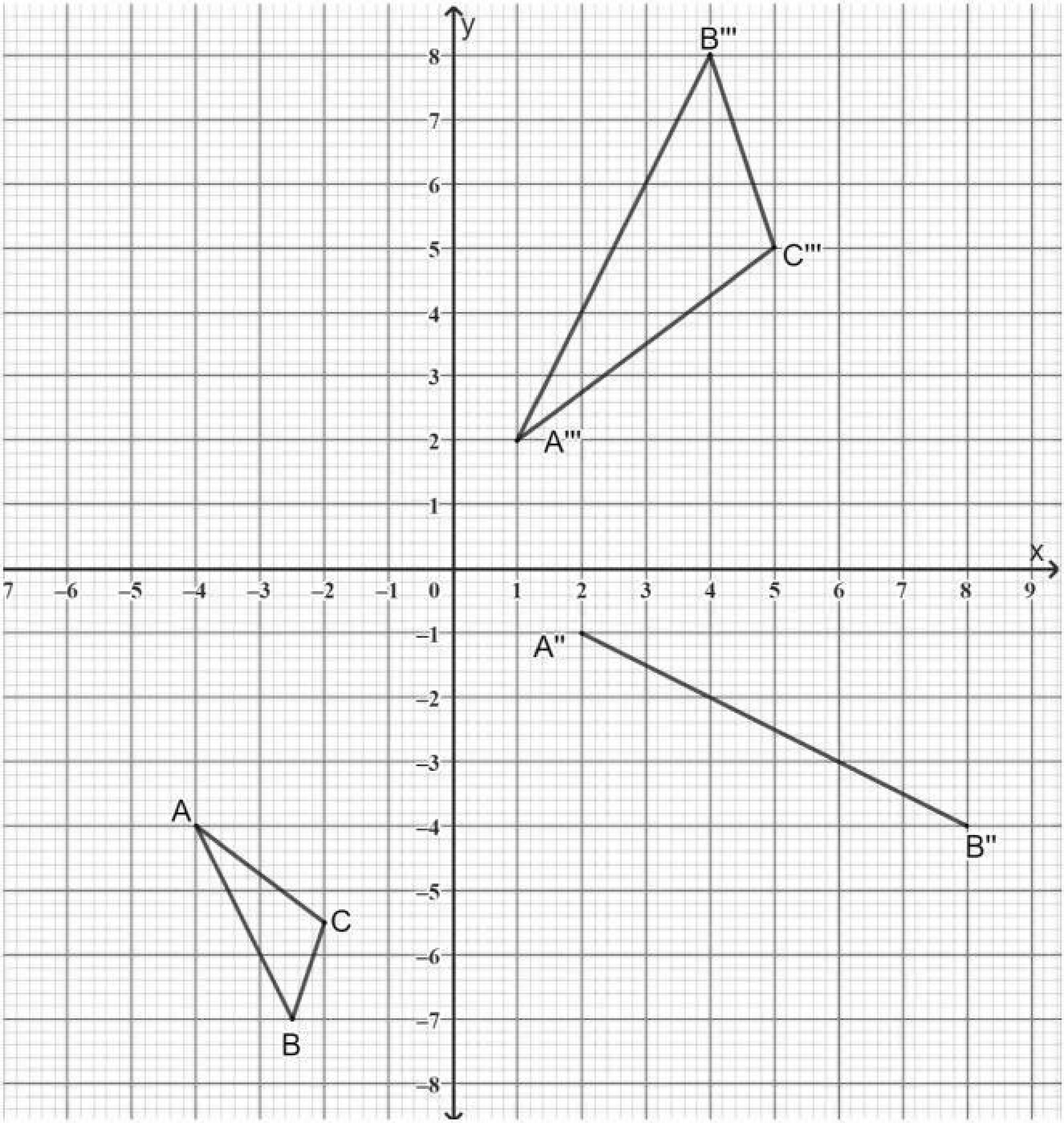
(b) Determine the median mark. (1 mark)

(c) State the mode of the data. (1 mark)

12. Use the cartesian plane provided to solve graphically the simultaneous equations $2x + 3y = 1.2$ and $5x + 4y = 1$



13. On the cartesian plane below, triangle ABC has vertices A (–4, –4), B (–2.5, 7) and C (–2, –5.5) while triangle A"B"C" has vertices A" (1, 2), B" (4, 8) and C" (5, 5). The line joining A" (2, –1), B" (8, –4) is part of another triangle A" B" C".



(a) Triangle A' B' C' is the image of triangle ABC under an enlargement centre (−3, −2) and scale factor (S.F.) = −2.

On the same grid, draw ΔA' B' C'. (3 marks)

(b) Triangle A'' B'' C'' is the image of triangle A' B' C' under rotation centre O (0, 0)

(i) State the angle of rotation. (1 mark)

(ii) Complete triangle A'' B'' C'' (2 marks)

(c) Triangle A" B" C" is the image of triangle A' B' C' under a reflection.

(i) Draw the mirror line. (1 mark)

(ii) Determine the equation of the mirror line in the form $y = mx + c$ (2 marks)

(d) Describe fully a single transformation that maps triangle

A' B' C' onto triangle A" B" C" (1 mark)

14. John cycles from shopping centre A on a bearing of 120° for 5 km to shopping centre B. He then cycles on a bearing of 200° for 7 km to the shopping centre C. Calculate to 1 decimal place.

a. The direct distance from A to C.

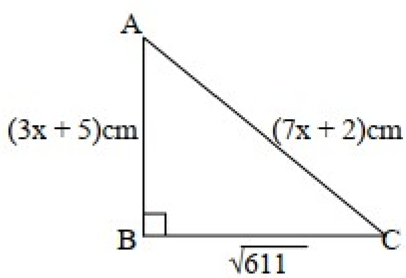
b. The bearing of A from C.
c. Bearing of B from C.

15. Given that tan x = $^5/_{12}$, find the value of the following without using mathematical tables or calculator:

a. Cos x

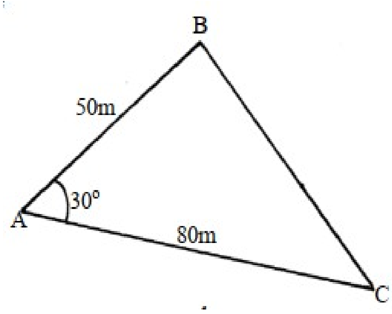
b. $\text{Sin}^2(90\text{-}x)$

16. In a triangle ABC, angle B is 90°. Find the value of x and hence the area of the triangle



17. Solve the following inequalities and represent the solution on a number line hence state the integral values $7x - 4 \leq 9x + 2 < 3x + 14$

18. The figure below represents a triangular plot ABC. The lengths of AB = 50m, AC = 80m and angle BAC =30°



a. Find the length of BC to 2 s.f

b. Find the area of the plot in hectares

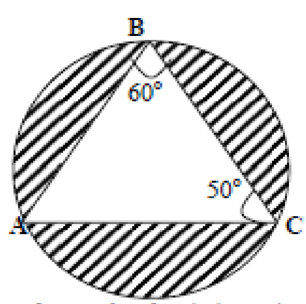
c. The plot is fenced using 4 strands of barbed wire. The length of one roll of barbed wire is 600m and it costs shs.4000. Calculate;

i. The length of fencing wire required

ii. The number of complete rolls to be bought

iii. The cost of the rolls

19. The figure below is a circle of radius 5cm. Points A, B and C are the vertices of the triangle ABC in which <ABC = 60° and <ACB=50° which is in the circle. Calculate the area of triangle ABC)



SERIES 2

Answer ALL questions in this section.

1. Use mathematical tables to find the reciprocal of 0.0025. (3 marks)

2. A sportsman named **Kibet** won Ksh 1,200,000 in a marathon. Write the amount in index form with base 10. (2 marks)

3. Simplify, leaving your answer in index form: (2 marks)
$3^{18} \div 3^{14}$

4. Solve for x: (2 marks)
$32^{x-1} = 64^{x}$

5. Work out: (2 marks)
$\log_2 32$

6. The height of a tower is 7.5 m. Use logarithm tables to find $\log_{10} 7.5$. (3 marks)

7. Use logarithm tables to evaluate: (3 marks)

$$3\sqrt{\frac{0.0452 \times 263}{1.456}}$$

8. A classroom floor has an area of $6x^2+17x+12)m^2$. Factorise the expression to determine its possible length and width. (3 marks)

9. A prime number is divisible only by 1 and itself. List all the prime numbers between 60 and 70. (3 marks)

10. Amina stored 210 kg of maize, 385 kg of beans, 150 kg of rice and 25 kg of millet. Classify the total mass as prime or composite. (2 marks)

11. Express $\frac{98}{14}$ in its simplest form. (1 mark)

12. Triangle **DEF** is similar to triangle **XYZ**.

In triangle DEF:

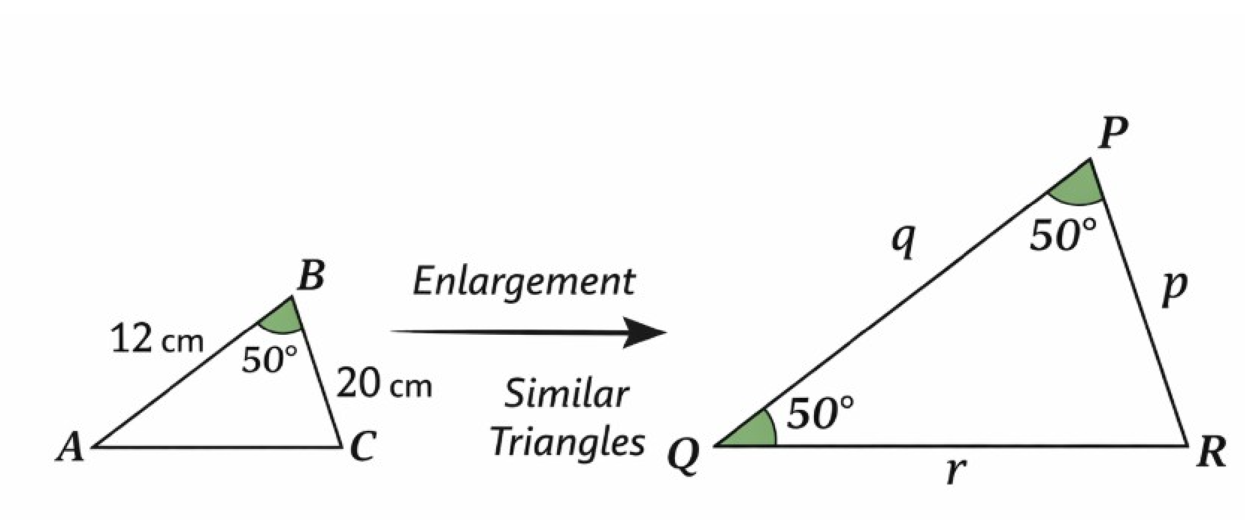
$\angle D = 40^\circ, \angle E = 60^\circ, \angle F = 80^\circ$

Side $DE = 6$ cm, $DF = 9$ cm

In triangle XYZ:

$\angle X = 40^\circ, \angle Y = 60^\circ, \angle Z = 80^\circ$

Side $XY = 12$ cm



Find the length of side XZ in triangle XYZ.(2marks)

13. Express the reciprocal of $\frac{9}{25}$ as a decimal. (2 marks)

14. Zawadi travelled 512 km in 8 hours to visit relatives.
Express distance and time as products of prime factors and use division law of indices to find speed. Leave answer in index form. (5 marks)

15. Solve for x and y: (5 marks)

$2^{3x} \div 2^{y} = 256$
$34^{x} \times 9^{y} = 81$

16. Mutiso owns a rectangular farm with area $(x^2 - 6x + 9)$ m². Factorise to find the length of one side. (5 marks)

17. A bus covered 360 km at a speed of v km/h. If it travelled 15 km/h faster, it would take 1.5 hours less.
Find the original speed. (5 marks)

18. The teacher wrote a certain mathematical exercise on the interactive board. Use it to answer question 18. (5 marks)

Show that triangle ABC in Fig. 11.23 is similar to triangle LMN in Fig. 11.22 below. Hence find the values of x and y.

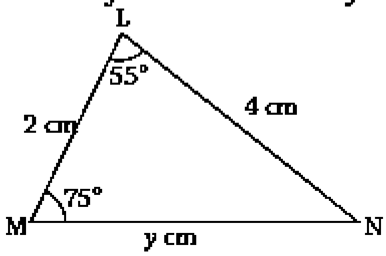


Fig. 11.22

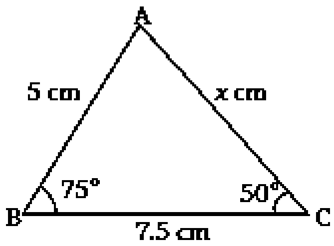


Fig. 11.23

19. On a graph, triangle ABC has coordinates A(2,2), B(4,4), C(6,2).
Find the coordinates after enlargement scale factor 2 about centre (1,1). (10 marks)

20. A poster is enlarged from 10 cm to 25 cm.
The larger poster has area 1250 cm^2.

a) Find the area scale factor. (3 marks)

b) Find the area of the smaller poster. (2 marks)

21. List all prime factors of:
a) 144 (2 marks)

b) 90 (2 marks)

c) 343 (2 marks)

d) 1331 (2 marks)

e) 5000 (2 marks)

22. Simplify and classify as rational or irrational:
a) $\sqrt{2.25}$ (2 marks)

b) $\sqrt{500}$ (2 marks)

c) $\sqrt{18}$ (2 marks)

d) $\sqrt{45}$ (2 marks)

e) $\sqrt{0.4096}$ (2 marks)

23. Use mathematical tables to find reciprocals:

a) 5.89 (3 marks)

b) 999 (3 marks)

c) 0.0000043 (4 marks)

SERIES 3

Answer ALL questions in this section.

1. Use mathematical tables to find the reciprocal of 0.0025. (3 marks)

2. A sportsman named **Kibet** won Ksh 1,200,000 in a marathon. Write the amount in index form with base 10. (2 marks)

3. Simplify, leaving your answer in index form: (2 marks)
$3^{18} \div 3^{14}$

4. Solve for x: (2 marks)
$32^{x-1} = 64^{x}$

5. Work out: (2 marks)
$\log_2 32$

6. The height of a tower is 7.5 m. Use logarithm tables to find $\log_{10} 7.5$. (3 marks)

7. Use logarithm tables to evaluate: (3 marks)

$$3\sqrt{\frac{0.0452 \times 263}{1.456}}$$

8. A classroom floor has an area of $6x^2+17x+12)m^2$. Factorise the expression to determine its possible length and width. (3 marks)

9. A prime number is divisible only by 1 and itself. List all the prime numbers between 60 and 70. (3 marks)

10. Amina stored 210 kg of maize, 385 kg of beans, 150 kg of rice and 25 kg of millet. Classify the total mass as prime or composite. (2 marks)

11. Express $\frac{98}{14}$ in its simplest form. (1 mark)

12. Triangle **DEF** is similar to triangle **XYZ**.

In triangle DEF:

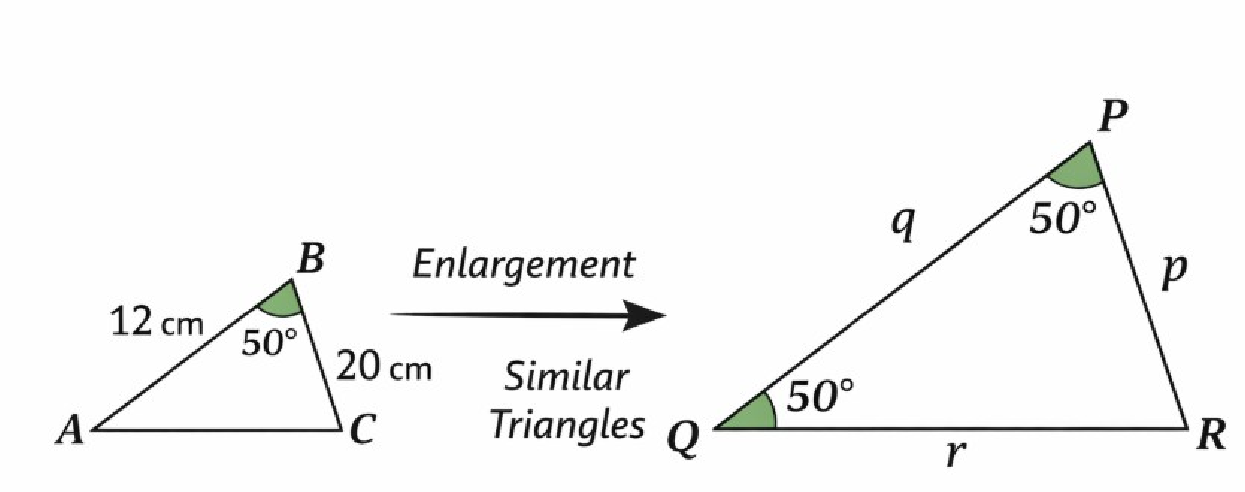
$\angle D = 40^\circ, \angle E = 60^\circ, \angle F = 80^\circ$

Side $DE = 6$ cm, $DF = 9$ cm

In triangle XYZ:

$\angle X = 40^\circ, \angle Y = 60^\circ, \angle Z = 80^\circ$

Side $XY = 12$ cm



Find the length of side XZ in triangle XYZ.(2marks)

13. Express the reciprocal of $\frac{9}{25}$ as a decimal. (2 marks)

14. Zawadi travelled 512 km in 8 hours to visit relatives.
Express distance and time as products of prime factors and use division law of indices to find speed. Leave answer in index form. (5 marks)

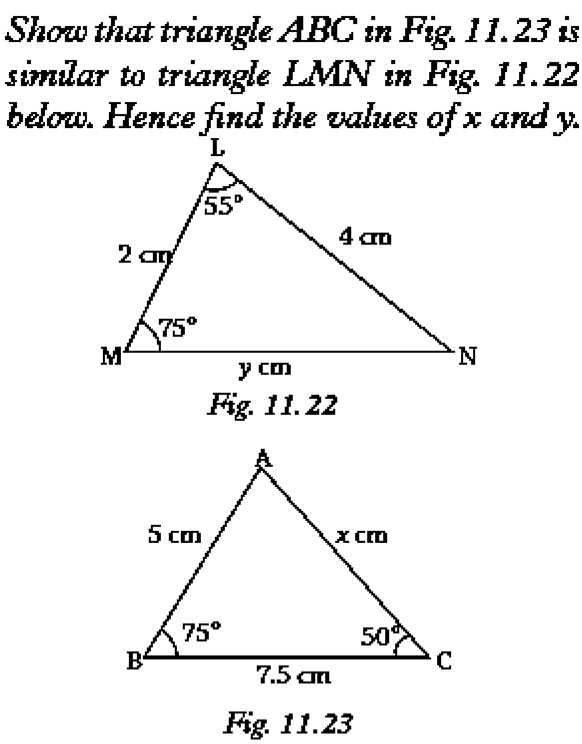
15. Solve for x and y: (5 marks)

$2^{3x} \div 2^{y} = 256$
$34^{x} \times 9^{y} = 81$

16. Mutiso owns a rectangular farm with area $(x^2 - 6x + 9)$ m². Factorise to find the length of one side. (5 marks)

17. A bus covered 360 km at a speed of v km/h. If it travelled 15 km/h faster, it would take 1.5 hours less. Find the original speed. (5 marks)

18. The teacher wrote a certain mathematical exercise on the interactive board. Use it to answer question 18. (5 marks)



19. On a graph, triangle ABC has coordinates A(2,2), B(4,4), C(6,2).
Find the coordinates after enlargement scale factor 2 about centre (1,1). (10 marks)

20. A poster is enlarged from 10 cm to 25 cm.
The larger poster has area 1250 cm².

a) Find the area scale factor. (3 marks)

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c) $\sqrt{18}$ (2 marks)

d) $\sqrt{45}$ (2 marks)

e) $\sqrt{0.4096}$ (2 marks)

23. Use mathematical tables to find reciprocals:
a) 5.89 (3 marks)

b) 999 (3 marks)

c) 0.0000043 (4 marks)

SERIES 4

SECTION A: 40 MARKS

1. Simplify: $(2x^2y)(3xy^3)$[4 marks]

2. Solve the equation: $3(2x - 5) = 4x + 7$[4 marks]

3. A number is increased by 20% to give 96. Find the original number.[4 marks]

4. Evaluate: $2^3 \times 2^5 \div 2^4$[4 marks]

5. Factorize completely: $x^2 - 9x + 20$ [4 marks]

6. A triangle has sides 6 cm, 8 cm and 10 cm. Determine whether it is right-angled. Give a reason.

[4 marks]

7. Simplify the ratio 24:36:60[4 marks]

8. A machine gives a 15% loss. If the cost price is Sh 2,000, find the selling price.[4 marks]

9. The average of 6, 8, 10, x and 14 is 10. Find x.[4 marks],

10. A class has 12 boys and 18 girls. Find the probability of selecting a boy at random.[4 marks]

SECTION B: 60 MARKS

Answer any five questions from this section

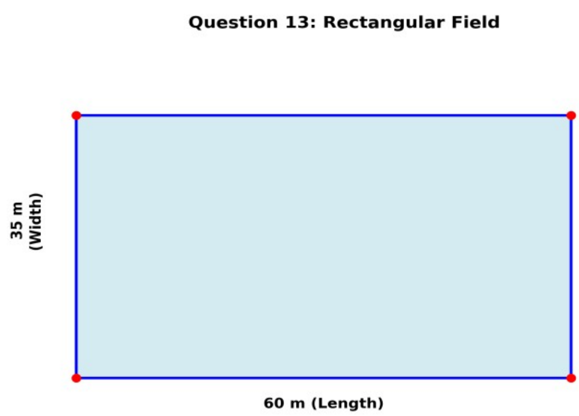
11. Solve the simultaneous equations and find $3x - 2y$
(a) Solve: $2x + y = 11$ and $x - y = 1$[4 marks]

(b) Hence find the value of $3x - 2y$[2 marks]

12. The coordinates of triangle PQR are P(1, 2), Q(5, 2) and R(5, 6).
(a) Plot the triangle on a Cartesian plane[2 marks]

(b) Find the area of triangle PQR[4 marks]

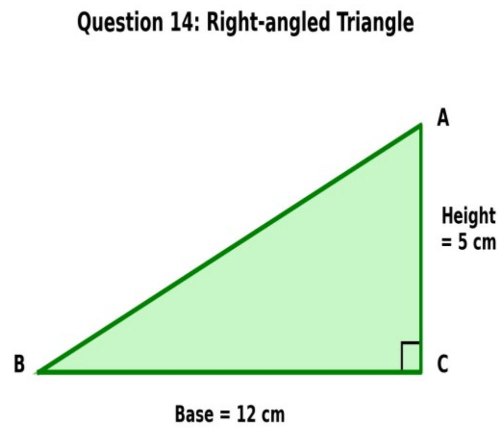
13. A rectangular field is 60 m long and 35 m wide.



(a) Find its perimeter.[2 marks]

(b) Find its area in m² and hectares.[4 marks]

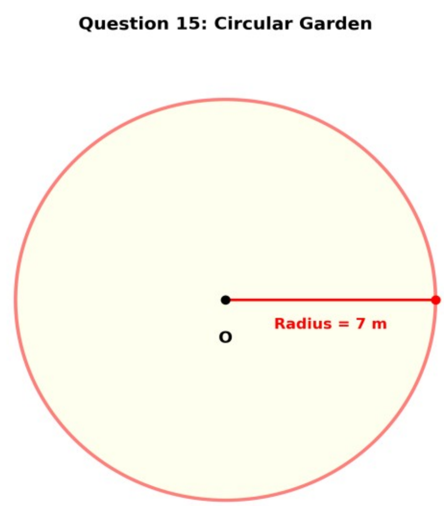
14. A right-angled triangle has base 12 cm and height 5 cm.



(a) Find its area.[2 marks]

(b) Find the length of the hypotenuse.[4 marks]

15. A circular garden has radius 7 m.



(a) Find its circumference.[2 marks]

(b) Find its area. (Use $\pi = 22/7$)[4 marks]

16. On a map, 1 cm represents 5 km. Two towns are 8.4 cm apart on the map.
(a) Find the actual distance between the towns.[3 marks]

(b) If a lorry travels at 60 km/h, find the time taken to travel the distance.[3 marks]

17. A bag contains 5 red, 3 blue and 2 green balls.
(a) Find the probability of picking a blue ball.[2 marks]

(b) Find the probability of picking a red or green ball.[4 marks]

18. The following data shows the number of books read by students:
2, 3, 4, 4, 5, 6, 6, 6, 7

(a) Find the mode.[1 mark]

(b) Find the median.[2 marks]

(c) Find the mean.[3 marks]

SERIES 5

SECTION I (50 MARKS)

Answer ALL questions in this section.

1. Simplify: $\sqrt{48} + \sqrt{75} - \sqrt{108}$ **(3 marks)**

2. Solve for x: $4^{x+1} = 8^{x-1}$ **(3 marks)**

3. The 3rd and 7th terms of an arithmetic progression are 10 and 22 respectively. Find the first term and the common difference. **(3 marks)**

4. Factorize completely: $6x^2 - 7x - 3$ **(3 marks)**

5. Given that log 2 = 0.3010 and log 3 = 0.4771, evaluate without tables or a calculator: log 72 **(4 marks)**

6. In triangle PQR, PQ = 8 cm, PR = 6 cm and angle QPR = 120°. **(3 marks)**
Calculate the length QR, correct to 2 decimal places.

7. Evaluate: $\log_5 125 + \log_5 (1/25)$ **(2 marks)**

8. Simplify: $(3^{n+2} - 3^n) \div (2 \times 3^n)$ **(3 marks)**

9. Rationalize the denominator and simplify: $(2 + \sqrt{3}) / (\sqrt{3} - 1)$ **(4 marks)**

10. Solve the quadratic equation: $2x^2 - 7x + 3 = 0$ **(3 marks)**

11. In a geometric progression, the 2^{nd} term is 6 and the 5^{th} term is 162. Find the first term and the common ratio. **(3 marks)**

12. Form a quadratic equation with integer coefficients whose roots are $x = -2$ and $x = 5$. **(3 marks)**

13. Using the sine rule, find angle B in triangle ABC where $a = 7$ cm, $b = 9$ cm and angle A = 43°. Give your answer to the nearest degree. **(4 marks)**

14. A geometric progression has a first term of 3 and a sum to infinity of 12. Find the common ratio. **(3 marks)**

15. Expand and simplify: $(\sqrt{5} + 2)^2$ **(3 marks)**

16. The sum of the first n terms of a series is given by $S_n = 2n^2 + 3n$. Find the 5^{th} term of the series. **(3 marks)**

SECTION II (50 MARKS)

Answer any FIVE questions from this section. Each question carries 10 marks.

17.

(a) An arithmetic progression has first term 5 and common difference 3. **(10marks)**

(i) Write down the first four terms of the progression. **(1 mark)**

(ii) Find the 20^{th} term. **(2 marks)**

(iii) Calculate the sum of the first 20 terms. **(2 marks)**

(b) The sum of the first n terms of a series is $S_n = 3n^2 - n$.

(i) Find the first three terms of the series. **(2 marks)**

(ii) Show that the series is an arithmetic progression and state its common difference. **(3 marks)**

18. **(10 marks)**

(a) Find the sum of the geometric series $2 + 6 + 18 + \ldots$ to 8 terms. **(3 marks)**

(b) A geometric progression has terms 128, 64, 32, …

(i) Find the 7^{th} term. **(2 marks)**

(ii) Find the sum to infinity of the series. **(2 marks)**

(c) Three consecutive terms of a geometric progression are $(x - 2)$, $(x + 2)$ and $(3x + 3)$. Find the value of x. **(3 marks)**

19. **(10 marks)**

(a) Without using tables or a calculator, evaluate: $\log_2 64 - \log_2 16 + \log_2 4$ **(3 marks)**

(b) Solve for x: $\log (2x + 3) - \log (x - 1) = 1$ **(4 marks)**

(c) Given that $\log 5 = 0.6990$ and $\log 7 = 0.8451$, evaluate without using tables:

(i) $\log 35$ **(1 mark)**

(ii) log 175 **(2 marks)**

20.

(a) Simplify, leaving your answer as a power of 2: $(2^{n+3} \times 4^{n-1}) \div 8^{n}$ **(4 marks)**

(b) Rationalize the denominator and simplify: $(3\sqrt{2} + \sqrt{6}) / (\sqrt{2} - \sqrt{3})$ **(3 marks)**

(c) Solve the equation: $9^{x} - 10(3^{x}) + 9 = 0$ **(3 marks)**

21.

(a) Solve $x^2 - 6x + 7 = 0$ by completing the square, giving answers correct to 2 decimal places. **(4 marks)**

(b) Find the range of values of x for which: $2x^2 + 3x - 14 \leq 0$ **(3 marks)**

(c) The sum of a number and its reciprocal is 29/10. Find the possible values of the number. **(3 marks)**

22.

(a) In triangle ABC, AB = 7 cm, BC = 5 cm and angle ABC = 115°.

(i) Calculate the length AC, correct to 1 decimal place. **(3 marks)**

(ii) Find angle BAC, correct to the nearest degree. **(3 marks)**

(b) In triangle PQR, angle P = 54°, angle Q = 72° and PQ = 10 cm.
Calculate the length PR, correct to 2 decimal places.

23.

(a) A rectangular garden has a perimeter of 44 m and an area of 120 m².
Form a quadratic equation and use it to find the length and width of the garden.

(b) Solve the equation: $x^4 - 13x^2 + 36 = 0$ **(5 marks)**

24.

(a) Using logarithms (log 2 = 0.3010, log 3 = 0.4771), solve for x correct to 3 significant figures: $2^{2x+3} = 3^{x+1}$ **(5 marks)**

(b) A ball is dropped from a height of 16 m. Each time it bounces, it rises to 3/4 of the height of the previous bounce.

(i) Find the height reached after the 5th bounce. **(2 marks)**

(ii) Find the total distance the ball travels before coming to rest. **(3 marks)**

SERIES 6

SECTION A: (40 Marks)

Answer all questions in this section.

1. A shopkeeper sold 15 boxes of sugar. If the number of boxes sold yesterday was **15**, classify the number as **even, odd, prime, or composite**. Explain your answer.

2. A machine produces 8 units per hour. Find the **reciprocal** of the production rate.

3. Determine whether the following numbers are **rational or irrational**:

(a) $\sqrt{49}$

(b) $\sqrt{2}$

4. Using a calculator, find the reciprocal of 0.25 and show your working in fraction form.

5.Express $5^3 \times 5^4$ as a single power of 5.

6. Simplify $\frac{2^5 \times 2^3}{2^4}$.

7. Using logarithm tables, find $\log 250$ to **4 decimal places**.

8. Simplify $\log 100 + \log 0.1$ using the laws of logarithms.

9. Evaluate $(27)^{\frac{2}{3}}$.

10. Expand and simplify $(x + 3)(x - 5)$.

SECTION B: 60 MARKS

Answer any five questions.

11. A trader invests Ksh 5000 in a business. The growth of the investment is modeled by the expression $A = 5000 \times 2^n$, where n is the number of months.

a) Evaluate A after 3 months. (3 marks)

b) Express $2^3 \times 2^4$ as a single power and verify your answer. (3 marks)

c) Using logarithms, find the number of months n required for the investment to reach Ksh 80,000. (6 marks)

12. A group of students is designing a rectangular garden. The area A in square meters is given by $A = x^2 + 5x - 24$.

a) Factorize the quadratic expression. (4 marks)

b) Determine the possible values of x if $A = 0$. (4 marks)

c) Verify your solutions by substitution. (4 marks)

13. An engineer is analyzing load on a bridge. The load is modeled by $P = 10^x$.

a) If $P = 1000$, find x using logarithms to base 10. (4 marks)

b) Simplify $\frac{10^6 \times 10^2}{10^3}$ using the laws of indices. (4 marks)

c) Evaluate $(100)^{3/2}$. (4 marks)

14. A student measures the side lengths of a cube as x cm. The volume V is $V = x^3$.

a) Express V in terms of powers if $x = 2y$. (3 marks)

b) Simplify $(3^2)^4$ using the laws of indices. (3 marks)

c) If $V = 512$, find x. (6 marks)

15. A store reduces the price of a TV from Ksh 45,000 to Ksh 36,000.

a) Calculate the fraction representing the discount. (3 marks)

b) Express $45{,}000 - 36{,}000$ as a factorized quadratic if necessary. (3 marks)

c) If the discount is repeated every year, express the price after 3 years using powers. (6 marks)

16. The population of a town grows according to $P = 5000\,(1.1)^t$, where t is in years.

a) Find the population after 5 years. (4 marks)

b) Determine t if the population reaches 10,000 using logarithms. (4 marks)

c) Simplify $(1.1)^3 \times (1.1)^2$ using the laws of indices. (4 marks)

17. A rectangle has length $x + 2$ and width $x - 3$. The area is $2x^2 - 5x - 12$.

a) Form a quadratic equation from the area expression. (4 marks)

b) Solve the quadratic equation by factorization. (4 marks)

c) Interpret the solutions in the context of the rectangle's dimensions. (4 marks)

SERIES 7

Answer all questions provided in this assessment paper.

1. Find the value of $\boldsymbol{x}$ *(3marks)*

$$2^{(x-3)} \times 8^{(x+2)} = 128$$

2. Evaluate (3marks)

$$\frac{-12 \div (-3) \times 4 - (-15)}{-5 \times 6 \div 2 + (-5)}$$

3. Convert the recurring decimal $12.\dot{1}\dot{8}$ into fraction (3 marks)

4. Without using tables and calculators, evaluate (3marks)

$$\sqrt[3]{\frac{0.032+0.0608}{1.28\times 0.4}}$$

5. Use tables of squares, square roots and reciprocals to evaluate correct to 4 s.f (4marks)

$$\frac{3}{\sqrt{0.0136}} - \frac{2}{(3.72)^2}$$

6. Simplify as far as possible. (3marks)

$$\frac{3}{x-y} - \frac{1}{x+y}$$

7. Fifteen men working for eight hours a day can complete a certain job in exactly 24 days.For how many hours a day must sixteen men work in order to complete the same job in exactly 20 days. (3marks)

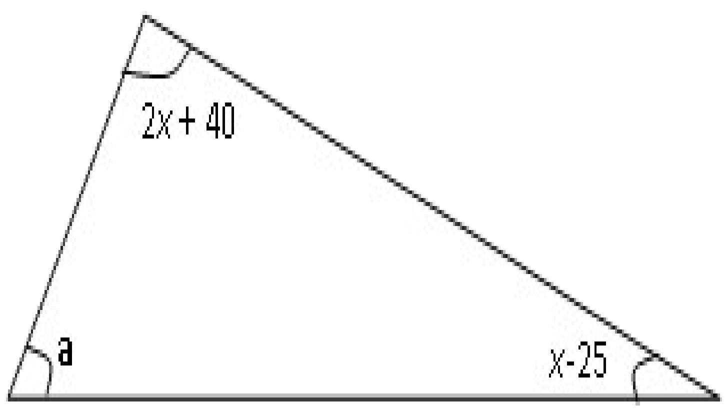
8. The scale of a map is 1:50000. A lake on the map is $6.16cm^2$. Find the actual area of the lake in hectares. (3marks)

9.Solve the inequality below and write down the integral values that satisfy the equality

$-3x + 2 < x + 6 \leq 17 - 2x$ (3 marks)

10. Find the equation of a line which passes through the point (2, 3) and is perpendicular to $y - 3x + 1 = 0$, giving your answer in the form $y = mx + c$ (3 marks)

11.In the figure below, angle **a** is half the sum of the other angles. Evaluate the triangle (3 marks)



12. Without using a protractor, draw a triangle ABC where $\angle CAB = 30°$, AC = 3.5cm and AB = 6cm. measure BC (3 marks)

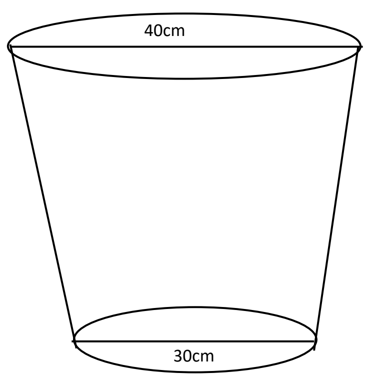
13. A solid cylinder of radius 6cm and height 12cm is melted and cast into spherical balls of radius 3cm. Find the number of balls made (3 marks)

14. The surface area of two similar bottles are $12cm^2$ and $108cm^2$ respectively. If the larger one has a volume of $810cm^3$. Find the volume of the smaller one (3 marks)

15. Evaluate using mathematical tables only. (3 marks)

$$\frac{6.373\log 4.948}{\sqrt{0.004636}}$$

16. The figure below shows a bucket of depth 30cm used to fill a cylindrical tank of radius 1.2m and height 1.35m which is initially three-fifth full of water.



Calculate, in terms of Π;

i)The capacity of the bucket in litres(3marks)

ii)The volume of water required to fill the tank in litres (2marks)

iii)Calculate the number of buckets that must be drawn to fill the tank (3marks)

17.The angle of depression of a point A on the ground from the top of a post is 18^0 and that of another point B on the same line as A nearer to the foot of the post is 25^0. If A and B are 70m apart,

(a) Draw a sketch to represent positions of A and B. (2marks)

(b) Using your sketch calculate

(i) The height of the post from the ground level (Ans 1 d.p) (4marks)

(ii) The distance of point A from the foot of the post.(2marks)

18.A field was surveyed and its measurements recorded in a field book as shown below.

	F	
	100	
E 40	80	
	60	D 50
C 40	40	
	20	B 30
	A	

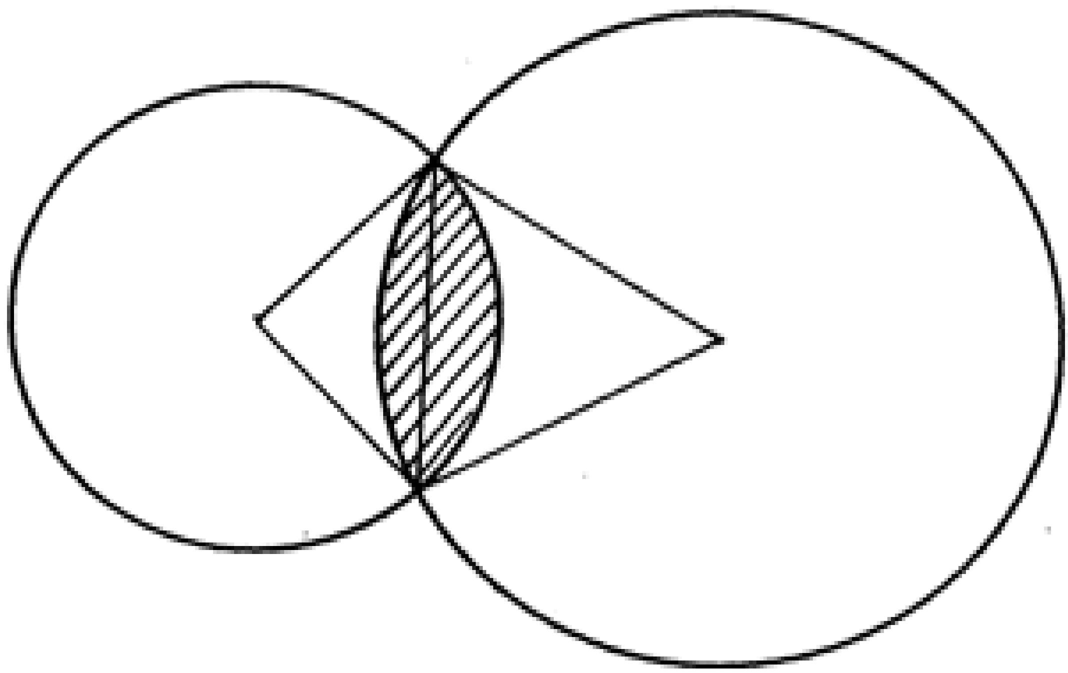
(a) Using a scale of 1cm to represent 10m, draw a map of the field. (4marks)

(b) Calculate the area of the field.

(i) in square metres. (4marks)

(ii) in hectares. (2marks)

19. The figure below shows two intersecting circles with centres P and Q and radius 5cm for the small one and 6cm for the big one. AB is a common chord of length 8cm. Calculate;



(a) the length of PQ (1 marks)

(b) the size of;

(i) angle APB (2marks)

(ii) angle AQB (2 marks)

(c) the area of the shaded region(4 marks)

20. Triangle PQR has vertices P(3,2), Q(-1,1) and R(-3,-1).

(a) Draw PQR on the grid provided. (1mark)

(b) Under a rotation the vertices of $P^1Q^1R^1$ are $P^1(1,4)$, $Q^1(2,0)$ and $R^1(4,-1)$. Find the centre and angle of rotation using points P and Q. (3marks)

(c) Triangle PQR is enlarged with scale factor 3 centreO(0,0) to give triangle $P^2Q^2R^2$. Draw triangle $P^2Q^2R^2$ and state its co-ordinates. (2marks)

(d) Triangle $P^1Q^1R^1$ undergoes reflection in line $y = -x$ to give triangle $P^3Q^3R^3$. Draw $P^3Q^3R^3$ and state its coordinates. (3marks)

21.Use a scale of 1cm represents 50km in these questions. Five towns **A**, **B**, **C**, **D** and **E** are situated such that **A** is 200 km from **B** on a bearing of 050° from **E**. **C** is 300 km from **B** on a bearing of 150° from **B**. **D** is 350km on a bearing of 240° from **C**. **E** is 200km from **D** and the bearing of **D** from **E** is 100°

a) Draw the diagram representing the positions of the towns (4marks)

b) From the diagram, determine;

i) The distance in km of A from E (3marks)

ii) The bearing of D from B (3marks)

SERIES 8

1. Define the following terms as used in mathematics and in each case list the numbers falling between 10 and 20

(a) Prime Number (2 marks)

(b) Composite Number (2 marks)

(c) Reciprocal (2 marks)

2. Solve the following quadratic equation using the completing square method (4marks)

   $X^2 +3x-8=0$

3. Using the calculator find the log of 2345 (2 marks)

4. Write down the quadratic identities for:

   (a) $(a+b)^2$

   (b) $(a-b)^2$ (2 marks)

5. By finding the logarithm of 0.00003567nusing the mathematical table,differentiate between characteristic and mantissa (4 marks)

6. State the relationship between the Linear Scale Factor (L.S.F) and the Area Scale Factor (A.S.F). (2 marks)

7. (a) Use mathematical tables to find the reciprocal of 0.0456. (3 marks)

(b) Using a calculator, determine the reciprocal of 125 and verify that the product of the number and its reciprocal is 1. (3 marks)

8. A vegetable garden measures 32m by 16 m. find its area and leave your answer as an index notation (3marks)

9. Using the logarithm tables find the value of (4marks)

$$\frac{3456 \times 0.000235 \times 1278}{0.0004567 \times 2398 \times 0.00897}$$

10. Expand the following equation (2x+3) (3x+4) (2marks)

11. Factorize the following quadratic equation (3marks)

12. Find the value of y in the following equation using the factorization method

$2y^2 + 12y = -16$ (4marks)

13. Using completing square technique find the value of (4marks)Type equation here.

$2x^2 + 12x + 16 = 2$

14. Solve for the value of x in the following equation

$$16^{2x+3} \times 8^{2x+1} = 256$$ (4marks)

SECTION B

ANSWER ALL THE QUESTIONS FROM THIS SECTION.

15. The length of two corresponding two sides of two similar cuboids is 4cm and 12 cm. find
    A. The linear scale factor (2marks)

    B. If the base area of the smaller one was $25cm^2$ find the base area of the larger cuboid (4marks)

    C. If the larger one had a mass of 600grams find the mass of the smaller one in kgs (4marks)

16. A triangle A (2,1) B (4,1) and C (3,5) is reflected on x axis to get a A'B'C'. The same triangle ABC is reflected on line y= -3x + 1. Draw both the object and images on the same cartesian plane plane on the graph paper provided (10marks)

17. A triangle with vertices F (3,1), G (5,1) and H (5,4) is enlarged using scale factor 3, with the Centre of enlargement as (2,1). Draw both the object and image after enlargement and state the co-ordinates of the F'G'H (10 marks)

18. Nathan purchased some books for shs 1200. If the cost of the book was 10 shillings less then the number of books he could have purchased could be 4 more. Taking the initial cost of a book as x find the number of books he could have bought (10marks)

19. Kinyus and kamau had 45 mangoes together. After losing 5 mangoes each the product of the number of mangoes they both have is 124. Find the number of mangoes they initially had. (10marks)